Project Proposal

You’re on a new funding committee for a program that helps ports cut emissions from shipping. Your job is to judge whether green methanol could work well as a ship fuel. Green methanol is made by combining captured CO_2 with green H_2. The H_2 comes from splitting water using electricity, and that electricity needs to be low carbon. Using background sources, you will describe the main steps of the process, make a simple first estimate of how much material and energy it needs, and decide if the idea should get funding. Two things drive the project. CO_2 has to be captured, cleaned, and compressed before it can be used. Making H_2 also takes a lot of electricity. Because of that, both cost and emissions depend a lot on where the CO_2 and electricity come from.

- **Motivation:** Why is it hard to cut emissions from ocean shipping?

- **Innovation:** What steps are needed to clean and compress CO_2 before it can be used, and why are those steps important?

- **Calculation:** How much CO_2, H_2, and energy are needed to make a set amount of green methanol? How much would the electricity cost over a reasonable range of prices? Which assumption matters the most, and which step is most likely to drive cost or energy use?

- **Realistic expectations:** What has to be true for this to actually cut emissions, especially around low carbon electricity and reliable CO_2 tracking? What tradeoffs show up for money, people, the environment, etc.?